

Macroeconomic Theory II, Spring 2025: Homework 3

The assignment is due Friday, 04/25.

Problem 1

Consider an income process $y_t = \exp(w_t)$ with

$$w_t = \bar{w} + \rho w_{t-1} + \varepsilon_t,$$

where $\rho = 0.90$, and ε_t is Normally distributed with mean zero and variance $\sigma_\varepsilon^2 = 0.06$.

Set $\bar{w}$ so that the income process in levels y_t has mean 1 always, independently of the value of the variance of w_t (Remark: you have to be careful because the mean of $\exp(w_t)$ will depend on the variance of ε_t).

a) Using your favorite programming language, write a program to approximate the above income process using the Tauchen method with a five-state Markov chain.

b) Repeat point a) using a eleven-state Markov chain. Do you see any substantial improvement in the accuracy of the approximation? Discuss.

c) Now use the Rouwenhorst method with five states. Compare it to the Tauchen's approximation with 5 states that you calculated in point a) above. Can you argue that it performs better than Tauchen's method?

d) Repeat point c) with $\rho = 0.98$ and adjust σ_ε^2 so that w_t has the same standard deviation as in point c). Keep the mean of the process to zero.

Problem 2

Consider the consumption-saving problem discussed in class,

$$\begin{aligned} V(a, y) &= \max_{\{c, a'\}} \left\{ u(c) + \beta \sum_{y' \in Y} \pi(y', y) V(a', y') \right\} \\ &\quad s.t. \\ c + a' &\leq Ra + y \\ a' &\geq -\phi \end{aligned}$$

Assume that:

- The income process $y = \exp(w)$ follows the five-state Markov chain that you computed in Problem 1c with the Rouwenhorst method (i.e. with $\rho = 0.90$).

- The borrowing limit is $\phi = 0$, i.e., the agent is not allowed to accumulate any debt.
- Period utility is given by $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ with $\gamma = 2$, and the discount factor is $\beta = 0.95$.
- The interest rate is $r = 0.02$.

a) In your favorite programming language, compute the decision rule $a'(a, y)$, using at least two of the solution methods discussed in class.

b) By simulation, assess which of the two solution methods you used is more accurate.

c) Using the more accurate of the two solution methods, compute how the unconditional standard deviation of individual consumption varies across three different levels of risk aversion: $\gamma = 1, \gamma = 2, \gamma = 5$. Explain how the parameter γ controls the desire for consumption smoothing.

d) By changing the variance of the innovations to the income shock σ_ε (but keeping ρ unchanged) study how the saving-income ratio (the saving rate) varies as uncertainty rises. Report the saving rate for different values of $\sigma_\varepsilon^2 = \{0.01, 0.06, 0.12\}$. *NOTE: this requires solving Problem 1 each time you change the value for σ_ε^2 .*

e) Solve the model again, this time by imposing the natural borrowing limit $\phi = -y_{\min}/r$. Compare average consumption between the two economies with different debt limits. What can you conclude about the effect of the borrowing constraints on average consumption?

f) Define the innovation to income as $\varepsilon_t \equiv w_t - 0.90w_{t-1}$, where w_t and w_{t-1} are realizations of the Markov chain along the simulation. Define an *insurance coefficient* as

$$\psi = 1 - \frac{\text{cov}(\Delta \log c_t, \varepsilon_t)}{\text{var}(\varepsilon_t)},$$

i.e. as the fraction of the shock ε_t that does not translate into consumption growth $\Delta \log c_t = \log c_t - \log c_{t-1}$. Compute ψ by simulation for two cases: zero borrowing constraint and natural borrowing constraint. Argue that, based on the value of ψ , the model where the borrowing limit is looser displays more consumption insurance.